

Learning to realize education's promise

DEON FILMER and HALSEY ROGERS



SCHOOLING is not the same as learning. In Kenya, Tanzania, and Uganda, when grade 3 students were asked to read a simple sentence like, ‘The name of the dog is Puppy’, three-quarters did not understand what it said. In rural India, nearly three-quarters of students in grade 3 could not solve a two-digit subtraction such as 46-17 – and by grade 5 half could still not do so. Although the skills of Brazilian 15-year-olds have improved, if they continue to improve at their current rate they will not reach the rich country average score in math for 75 years. In reading, it will take 263 years. These are all countries that have measured learning and made the results public; in too many other countries the problem remains hidden.

Schooling without learning is not just a wasted opportunity, but also a great injustice: the children society is failing the most are the ones in greatest need of a good education to succeed in life. Without learning, education fails to deliver fully on its promise as a driver of poverty elimination and shared prosperity. Within countries, learning outcomes are almost always much worse for the disadvantaged. In Uruguay, poor children in grade 6 are assessed as ‘not competent’ in math at five times the rate of wealthy children. Moreover, these results are for children and youth lucky enough to be in school. Many aren’t even enrolled in primary or secondary school, with members of disadvantaged groups – poor children, girls, children with disabilities, ethnic minorities – most likely to be out of school. Together, these severe shortfalls constitute a *learning crisis*.

Education should equip students with the skills they need to lead healthy, productive, meaningful lives. Different countries will define skills differently, but all share some core aspirations, embodied in their curriculums. Students everywhere must learn how to interpret many types of written passages – from medication labels to job offers, from bank statements to great literature. They have to understand how numbers work so that they can make transactions in markets, set family budgets, interpret loan agreements, or write engineering software. They require the higher order reasoning and creativity that builds on these foundational skills. And they need the socio-emotional skills – like perseverance and the ability to work on teams – that help them apply what they have learned. But the learning that one would expect to happen in schools – whether expectations are based on formal curriculums, needs of employers, or just common sense – is often not occurring, especially for the already marginalized.

Learning shortfalls during the school years eventually show up as weak skills in the workforce – so when people in countries around the world debate the job-skills problem, they’re really talking about the learning crisis. Because education systems have not prepared workers adequately, many enter the labour force with inadequate skills. Recent initiatives have assessed a range of skills in the adult populations of numerous countries. They find that even foundational skills such as literacy and numeracy are often low, let alone more advanced skills. The problem is not just a lack of trained workers; it is a lack of readily trainable workers. Accordingly, many workers end up in jobs that require minimal amounts of reading or math. A lack of skills reduces job quality, earnings and labour mobility.

The good news is that there is nothing inevitable about low learning in low and middle income countries: when improving learning is a priority, great progress is possible. Starting in the early 1950s as a war-torn society with very low literacy

rates, the Republic of Korea had by 1995 achieved universal enrolment in high quality education through secondary school. In fact, its young people were performing at the highest levels on international learning assessments. Vietnam surprised the world when the 2012 results of the Programme for International Student Assessment (PISA) showed that its 15-year-olds were performing at the same level as Germany's, despite Vietnam's much lower income. Between 2009 and 2015, Peru achieved some of the fastest growth in overall learning outcomes due to concerted policy action and system reform. And in Liberia, Papua New Guinea, and Tonga, early grade reading improved substantially within a short time thanks to focused evidence based efforts.

But tackling the learning crisis requires first understanding what causes it. Around the world, the crisis has three main dimensions. The first is the poor learning outcomes themselves. As described earlier, *levels of learning are low*, and not just in the poorest countries. Many high performing students in some middle income countries (such as Algeria, the Dominican Republic, or Kosovo) would rank in the bottom quarter of students in the average Organization for Economic Co-operation and Development (OECD) country.

At the same time, *inequalities in learning outcomes are high*. By the end of primary school, only 5% of girls in Cameroon from the poorest quintile of households have learned enough to continue school, compared with 76% of girls from the richest quintile. And *improvements in system-wide learning are often slow*. In fact, across all countries participating in multiple rounds of the PISA since 2003, the median gain in the national average score from one round to the next was zero.

The second dimension of the learning crisis is its immediate causes, visible in schools in the various ways in which the teaching-learning relationship breaks down. The *World Development Report 2018* (WDR 2018) identifies four key ways this happens:

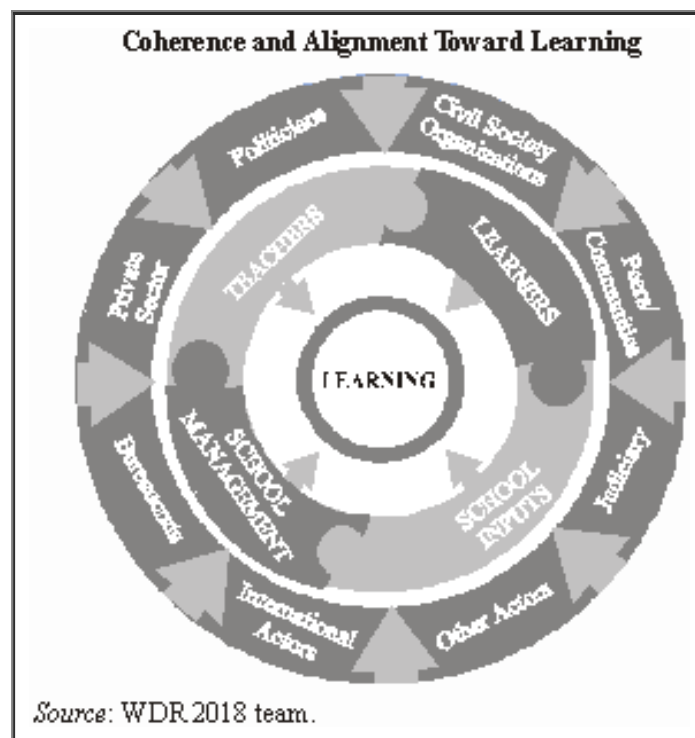
1. Because of poverty, many children arrive at school unprepared to learn. Malnutrition, illness, low parental investments, and the harsh environments associated with poverty undermine early childhood learning. Thirty per cent of children under 5 in developing countries are physically stunted – that is, they have low height for their age, typically due to chronic malnutrition. Poor developmental foundations mean that many children arrive at school unprepared to benefit fully from it, with poor children's cognitive skills falling well behind in the years before primary school.

In some countries, the gap between richer and poorer children's ability to recognize letters of the alphabet doubles between the ages of 3 and 5. Moreover, the fees and opportunity costs of schooling keep many young people out of school, and the social dimensions of exclusion – for example, barriers linked to gender or disability – exacerbate the problem. These inequalities in school participation widen the gaps in learning outcomes.

2. Teachers often lack the skills or motivation to teach effectively. Teachers are the most important factor affecting learning in schools. In the United States, students with great teachers learn three times as fast as those with poor teachers; in developing countries, teacher quality can matter even more. But most education systems do not attract applicants with strong backgrounds or train teachers effectively. For example, 15-year-old students who aspire to be teachers score below average on the PISA in nearly all countries. Across 14 Sub-Saharan countries, the average grade 6 teacher performs no better on reading tests than the highest performing grade 6 students. Moreover, much learning time is lost because classroom time is spent on other activities or teachers are absent. In seven Sub-Saharan countries, one in five teachers was absent from school during recent

unannounced visits by survey teams, with another fifth of teachers at school but absent from the classroom. These diagnostics are not intended to blame teachers; rather, they call attention to how education systems undermine learning by failing to support their frontline educators.

3. Inputs often fail to reach classrooms or to affect learning. Public discourse often equates problems of education quality with input gaps. Devoting enough resources to education is crucial, but input shortages explain only a small part of the learning crisis. One reason is that inputs often fail to make it to the frontlines. In Sierra Leone, for example, textbooks were distributed to schools, but follow up inspections found most of them locked away in cupboards, unused. Similarly, many technological interventions fail before they reach classrooms, and even when they do make it to classrooms, often do not enhance teaching or learning. In Brazil, the One Laptop Per Child initiative in several states faced years of delays. Then, even a year after the laptops finally made it to classrooms, more than 40% of teachers reported never or rarely using them in classroom activities.



4. Poor management and governance often undermine schooling quality. Although effective school leadership does not improve student learning directly, it does so indirectly by strengthening teaching and ensuring effective use of resources. Careful analysis of management practices shows that schools in developing countries often suffer from poor management, compared with schools in richer countries or even with manufacturing firms in their own countries. Ineffective school leadership means that school principals are not actively involved in helping teachers solve problems, do not provide instructional advice, and do not set goals that prioritize learning. Moreover, in many settings schools lack any meaningful autonomy, and community engagement fails to affect what happens in classrooms.

The third dimension of the crisis is its deeper systemic causes. All these breakdowns in schools and communities are driven by deeper system level factors – often invisible – that pull key actors away from a focus on learning. For one thing, operating an education system effectively poses major *technical* challenges: parts of the system need to be aligned with learning and coherent with each other, and actors at all levels need the capacity to implement well. But many of the deeper causes of the learning crisis are *political* in nature. Actors have interests that diverge from learning. Politicians act to preserve their positions of power, which may lead them to target certain groups (geographic, ethnic, or economic) for benefits. Bureaucrats may focus more on keeping politicians and teachers happy

than on promoting student learning, or they may simply try to protect their own positions.

Private suppliers of education services – whether textbooks, construction, or schooling – may, in the pursuit of profit, advocate policy choices that undermine learning. Teachers and other education professionals, even when motivated by a sense of mission, also may fight to maintain secure employment and to protect their incomes. None of this is to say that education actors don't care about learning – rather, especially in poorly managed systems, competing interests may loom larger than the learning-aligned interests. These technical and political challenges keep many systems stuck in low-learning traps, with low accountability and high inequality.

To do better, a nation must assess learning, to make it a serious goal; act on evidence, to make schools work for learners, and align actors, to make the system work for learning.

First, assess learning, to make it a serious goal. Only half of all countries have metrics to measure learning at the end of primary and lower secondary school – indicators that are required to monitor progress toward the United Nations' Sustainable Development Goals for learning. Fewer still have the ability to track learning over time.

Countries need to put in place a range of well designed student assessments to help teachers guide students, improve system management, and focus society's attention on learning. These measures can put the spotlight on hidden exclusions, inform policy choices, and track progress.

Second, act on evidence, to make schools work for learners. Great schools are those that build strong teaching-learning relationships in the classroom. Thanks to innovations by educators and advances in brain science, knowledge of how students learn most effectively has exploded. But common practice in schools and communities often diverges sharply from what evidence identifies as most promising.

Countries can use the evidence to close this gap and make schools work better. The best place to start is in these three key areas.

Prepared learners: Reduce stunting and promote brain development through early nutrition and stimulation (as in Chile) so children can learn. Support disadvantaged children with grants to keep them in school.

Skilled, motivated teachers: Attract talented people into teaching (as in Finland). Use repeated, specific teacher training reinforced by mentors (as in some African settings) instead of the ineffective one-off methods that are more common.

Inputs and management focused on teaching and learning: Deploy technologies that help teachers teach to the level of the student (as in Delhi). Strengthen the capacity and powers of school management (as in Indonesia), including principals.

Third, align actors, to make the whole system work for learning. All this innovation in classrooms is not likely to have much of an impact if system level technical and political factors prevent an emphasis on learning. When key actors are focused on non-learning goals (such as political or personal gain) or lack implementation capacity, even well designed innovations can't be scaled up or sustained.

Countries can escape low-learning traps by acting on three fronts as they implement reforms: (i) Deploy information and metrics to make learning politically

salient (as the NGO-led ASER and Uwezo programmes have done in India and East Africa); (ii) Build coalitions to shift political incentives toward learning for all (as Chile did early in its decades-long education reforms, or as Malaysia and Tanzania did recently with their collaborative societywide ‘labs’ to design reform programmes); and (iii) Use innovative and adaptive approaches to find out which approaches work best in their context (as Burundi did while rebuilding its education sector after conflict).

Together, these three policy actions can deliver a system in which the elements are coherent with each other and all elements align with learning (see figure).

Coherence and alignment toward learning are necessary to ensure that a society’s investments in education pay-off fully. Given the many returns to education – financial and nonfinancial, for both individuals and societies – some countries clearly need to invest more in education, especially as more youth complete primary and secondary school and go on to higher education. At the same time, virtually all countries should spend more effectively. Across countries, the relationship between public spending on education and learning outcomes is weak, but the right investments can pay off. Used in conjunction with the actions outlined here, financing for education can help pull systems out of low-learning traps and expand opportunity.

The pay-off to these efforts is education that delivers for growth and development. Delivered well, education cures a host of societal ills. For individuals, it promotes employment, earnings, health and poverty reduction. For societies, it drives long-term economic growth, spurs innovation, strengthens institutions and fosters social cohesion. But as the mounting evidence shows, it is the *skills* acquired through education, rather than just the years spent in school, that drive growth and equip individuals for work and life.

Reforms to strengthen learning will allow countries to reap the many benefits of education. For example, even a relatively modest improvement in learning – one that lifts all students to the level of the average student in Brazil – could increase long-term growth rates in a middle income country like Mexico or Turkey by around two percentage points. Rapid technological change makes these foundational skills even more important because it is those skills that enable workers and citizens to adapt rapidly to new opportunities. Countries have already made a start by getting so many children and youth into school. Now it is time to realize education’s promise by accelerating learning for all.

* This article is drawn from the *World Development Report 2018: Learning to Realize Education’s Promise*. Citations for the figures and assertions in the article can be found in the report, which is available at www.worldbank.org/wdr2018.

